

Chemical Bonding and Molecular Structure

Complete chapter notes — Lewis structures and formal charge, bond parameters, VSEPR shapes, hybridisation, molecular orbital theory worked for every diatomic, and hydrogen bonding.

The spine of this chapter: atoms bond to lower their energy, and there are exactly three theories, each answering a different question. **Lewis/VSEPR** answers *what shape?* **Valence bond theory (hybridisation)** answers *which orbitals overlap, and how strong is the bond?* **Molecular orbital theory** answers *how many bonds, and is it magnetic?* When a question mentions **paramagnetism or bond order**, it is asking for MOT; when it mentions **shape or bond angle**, it is asking for VSEPR.

1 · The Kössel-Lewis Approach

Atoms combine to attain the stable **octet** of the nearest noble gas — by transferring electrons (ionic bond) or by sharing them (covalent bond). A **Lewis symbol** shows the valence electrons as dots around the symbol; the number of dots gives the valence directly.

1.1 Writing a Lewis structure

METHOD 1 Drawing any Lewis structure

1. Count the **total valence electrons**. For an anion **add** one per unit of negative charge; for a cation **subtract** one per unit of positive charge.
2. Pick the **central atom** — the least electronegative one (never hydrogen, which forms only one bond).
3. Join each outer atom to the centre with a single bond (2 electrons each).
4. Distribute the remaining electrons as **lone pairs** on the outer atoms first, to complete their octets.
5. If the central atom is still short of an octet, convert lone pairs on the outer atoms into **multiple bonds**.
6. Check with **formal charges**; the best structure has formal charges closest to zero, with any negative charge on the most electronegative atom.

For CO_3^{2-} : $4 + 3(6) + 2 = 24$ valence electrons, C central, one C=O and two C-O⁻, giving three equivalent resonance structures.

Formal charge = (valence electrons in the free atom) – (lone-pair electrons) – $\frac{1}{2}$ (bonding electrons). Formal charges are a **bookkeeping device** for choosing between structures — they are not real charges on the atoms.

WORKED EXAMPLE 1 — FORMAL CHARGES IN OZONE

O₃ is drawn as O_{central} with one double and one single bond.

Central O: $6 - 2 - \frac{1}{2}(6) = +1$. Doubly bonded terminal O: $6 - 4 - \frac{1}{2}(4) = 0$. Singly bonded terminal O: $6 - 6 - \frac{1}{2}(2) = -1$.

Sum = $+1 + 0 - 1 = 0$, matching the neutral molecule — always check this.

1.2 Limitations of the octet rule

Failure	Explanation	Examples
Incomplete octet	Central atom has fewer than 8 electrons	LiCl, BeH ₂ , BCl ₃ (6 around B)
Expanded octet	Elements of period 3 onward have vacant d orbitals and may hold 10, 12 or more	PF ₅ (10), SF ₆ (12), H ₂ SO ₄ (12)
Odd-electron species	An odd total cannot be paired into octets	NO (11), NO ₂ (17)
Noble gas compounds	The octet was supposed to be inviolable	XeF ₂ , XeOF ₂ , KrF ₂

Beyond these, the rule says nothing about **molecular shape** and nothing about the **energy** (and hence the stability) of a molecule.

2 · The Ionic Bond

An ionic bond forms when one atom transfers electrons to another and the resulting oppositely charged ions are held by electrostatic attraction in a three-dimensional **lattice**.

Conditions that favour it: low ionization enthalpy of the metal, highly negative electron gain enthalpy of the non-metal, and — decisively — a **high lattice enthalpy**.

Lattice enthalpy is the energy required to completely separate one mole of a solid ionic compound into gaseous ions. For NaCl it is 788 kJ mol⁻¹. It rises with **higher ionic charges** and **smaller ionic radii** (roughly $\propto q_1q_2/r$), which is why MgO (2+, 2-) melts far above NaCl.

FREQUENTLY ASKED

The formation of Na^+ and Cl^- from the atoms is *endothermic* overall ($+496 - 349 = +147 \text{ kJ mol}^{-1}$). NaCl is stable only because the lattice enthalpy released on assembling the crystal far outweighs that cost. Ionic bonding is driven by the **lattice**, not by the ion formation.

2.1 Fajans' rules — covalent character in an ionic bond

Covalent character increases with (i) **small cation**, (ii) **large anion**, (iii) **high charge** on either ion, and (iv) a cation with a **pseudo-noble-gas (18-electron) configuration**, such as Cu^+ , Ag^+ , Zn^{2+} , which polarises better than an alkali-metal ion of the same size.

Hence LiCl is more covalent than NaCl ; AlCl_3 is more covalent than MgCl_2 ; and among the lithium halides, LiI is the most covalent. A more covalent halide is more soluble in organic solvents and has a lower melting point.

3 · Bond Parameters

Parameter	Definition	What to remember
Bond length	Equilibrium distance between the nuclei of two bonded atoms	Falls as bond order rises: $\text{C-C } 154 \text{ pm} > \text{C=C } 133 \text{ pm} > \text{C}\equiv\text{C } 120 \text{ pm}$
Bond angle	Angle between two bonds at an atom	Fixed by VSEPR; CH_4 109.5° , NH_3 107° , H_2O 104.5°
Bond enthalpy	Energy required to break one mole of bonds in the gaseous state	Rises with bond order: $\text{N}\equiv\text{N } 946 > \text{O}=\text{O } 498 > \text{H-H } 435.8 \text{ kJ mol}^{-1}$
Bond order	Number of bonds between two atoms	Higher order $\Rightarrow$ shorter bond, greater enthalpy, greater stability

For a polyatomic molecule such as water, breaking the two O-H bonds takes different energies (502 and 427 kJ mol^{-1}), so the tabulated value 464 is a **mean bond enthalpy**.

3.1 Resonance

When a single Lewis structure cannot describe a molecule adequately, the real structure is a **resonance hybrid** of several canonical forms. The hybrid is **lower in energy** — more stable — than any contributing structure.

In O_3 , both oxygen-oxygen bonds are **equal** at 128 pm — between a single bond (148 pm) and a double bond (121 pm). In CO_3^{2-} all three C-O bonds are identical, and in NO_3^- likewise, which no single Lewis structure can show.

THE CLASSIC MISCONCEPTION

1. The molecule does **not** flip between canonical forms, and it is **not** a mixture of them. It is a single structure that none of the drawings captures on its own.
2. Canonical forms have **no independent existence**; only the hybrid is real.
3. All canonical forms must have the **same positions of nuclei** — only electrons move. Moving an atom gives a different molecule, not a resonance form.

3.2 Polarity and dipole moment

$\mu = q \times d$, measured in **debye** ($1 \text{ D} = 3.33564 \times 10^{-30} \text{ C m}$). Dipole moment is a **vector**: the molecular dipole is the vector sum of the individual bond dipoles, so a molecule with polar bonds can still be non-polar.

Molecule	BeF ₂	CO ₂	BF ₃	CCl ₄	H ₂ O	NH ₃	NF ₃	HF
μ / D	0	0	0	0	1.85	1.47	0.24	1.78
Why	Symmetrical — bond dipoles cancel exactly				Unsymmetrical / lone pair adds to the dipole			

NH₃ VS NF₃ — THE SINGLE MOST ASKED COMPARISON

Both are pyramidal, yet $\mu(\text{NH}_3) = 1.47 \text{ D}$ while $\mu(\text{NF}_3)$ is only 0.24 D , even though N-F bonds are far more polar than N-H bonds.

Reason. In both molecules the nitrogen lone pair contributes a dipole pointing *away* from nitrogen, along the axis. In **NH₃** nitrogen is more electronegative than hydrogen, so the three bond dipoles point *towards* nitrogen — in the **same** direction as the lone-pair dipole, and the two **add**. In **NF₃** fluorine is more electronegative than nitrogen, so the bond dipoles point *away* from nitrogen — **opposing** the lone-pair moment, and they largely **cancel**.

4 · VSEPR Theory — Predicting Shape

Valence Shell Electron Pair Repulsion theory: the electron pairs in the valence shell of the central atom arrange themselves as far apart as possible, because like charges repel.

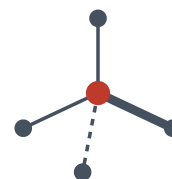
Repulsion order: lone pair-lone pair > lone pair-bond pair > bond pair-bond pair. A lone pair is held by only one nucleus, so it spreads out more and pushes harder.



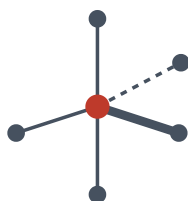
Linear · 180°
BeCl₂, CO₂



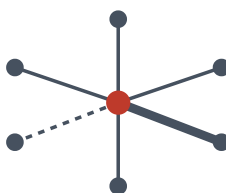
Trigonal planar · 120°
BF₃, BCl₃



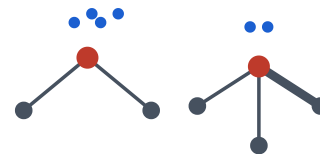
Tetrahedral · 109.5°
CH₄, NH₄⁺



Trigonal bipyramidal
PCl₅ · 90° and 120°



Octahedral · 90°
SF₆



Bent & pyramidal
H₂O 104.5° · NH₃ 107°

*The five basic geometries, plus the two commonest lone-pair shapes. Blue dots are lone pairs. The geometry is set by the total number of electron pairs; the **shape** is what you see when only the atoms are drawn.*

4.1 The complete VSEPR table

Type	Bond pairs	Lone pairs	Electron-pair geometry	Molecular shape	Example (angle)
AB ₂	2	0	Linear	Linear	BeCl ₂ , CO ₂ (180°)
AB ₃	3	0	Trigonal planar	Trigonal planar	BF ₃ (120°)
AB ₂ E	2	1	Trigonal planar	Bent	SO ₂ (119.5°), SnCl ₂
AB ₄	4	0	Tetrahedral	Tetrahedral	CH ₄ (109.5°)
AB ₃ E	3	1	Tetrahedral	Trigonal pyramidal	NH ₃ (107°)
AB ₂ E ₂	2	2	Tetrahedral	Bent	H ₂ O (104.5°)
AB ₅	5	0	Trigonal bipyramidal	Trigonal bipyramidal	PCl ₅ (90° , 120°)
AB ₄ E	4	1	Trigonal bipyramidal	See-saw	SF ₄
AB ₃ E ₂	3	2	Trigonal bipyramidal	T-shaped	ClF ₃
AB ₂ E ₃	2	3	Trigonal bipyramidal	Linear	XeF ₂ , I ₃ [−]
AB ₆	6	0	Octahedral	Octahedral	SF ₆ (90°)
AB ₅ E	5	1	Octahedral	Square pyramidal	BrF ₅
AB ₄ E ₂	4	2	Octahedral	Square planar	XeF ₄

MEMORY HOOK

In a **trigonal bipyramid**, lone pairs always take **equatorial** positions (only two neighbours at 90° instead of three), which is why SF_4 is a see-saw and ClF_3 is T-shaped. In an **octahedron** the second lone pair goes **trans** to the first, which is why XeF_4 is square planar rather than see-saw.

4.2 Bond-angle contraction

CH_4 (109.5°) > NH_3 (107°) > H_2O (104.5°). All three are sp^3 hybridised with four electron pairs; each lone pair added squeezes the bonding pairs closer together by about 2.5° .

Two further comparisons worth carrying: **NH_3 (107°) > NF_3 (102°)**, because the more electronegative fluorine pulls the bonding pairs away from nitrogen, reducing their mutual repulsion; and **NH_3 (107°) > PH_3 (93.6°)**, because the larger phosphorus atom uses nearly pure p orbitals, so the angle falls towards 90° .

5 · Valence Bond Theory and Hybridisation

VBT explains a covalent bond as the **overlap of two half-filled atomic orbitals** with opposite spins. Greater overlap gives a stronger bond, and the bond forms along the direction of maximum overlap — which is where directional character, and therefore shape, comes from.

5.1 Sigma and pi bonds

	σ (sigma) bond	π (pi) bond
Overlap	Head-on (axial): s-s, s-p, p-p along the axis	Sideways (lateral): p-p perpendicular to the axis
Strength	Stronger — overlap is larger	Weaker — overlap is smaller
Rotation	Free rotation possible about the bond	Restricted — this is the origin of geometrical isomerism
Occurrence	Always the first bond between two atoms	Only in addition to a σ bond

So a double bond is $1\sigma + 1\pi$, and a triple bond is $1\sigma + 2\pi$. In $\text{N}\equiv\text{N}$ (946 kJ mol^{-1}) the σ is formed by head-on $2p_z$ overlap and the two π bonds by sideways $2p_x$ and $2p_y$ overlap.

5.2 Hybridisation

Hybridisation is the intermixing of atomic orbitals of **nearly equal energy** belonging to the same atom, to give an equal number of **identical hybrid orbitals** of equal energy and shape. Hybrid orbitals are better directed and overlap more effectively, so they form stronger bonds.

Conditions: orbitals hybridised must belong to the same atom or ion, have comparable energies, and — importantly — hybridisation does *not* require the orbitals to be half filled; filled and empty orbitals may take part.

Hybridisation	Orbitals mixed	Geometry	Angle	s character	Examples
sp	1 s + 1 p	Linear	180°	50%	BeCl ₂ , C ₂ H ₂ , CO ₂
sp ²	1 s + 2 p	Trigonal planar	120°	33%	BF ₃ , C ₂ H ₄ , SO ₂
sp ³	1 s + 3 p	Tetrahedral	109.5°	25%	CH ₄ , NH ₃ , H ₂ O, NH ₄ ⁺
sp ³ d	1 s + 3 p + 1 d	Trigonal bipyramidal	90°, 120°	—	PCl ₅ , SF ₄ , ClF ₃
sp ³ d ²	1 s + 3 p + 2 d	Octahedral	90°	—	SF ₆ , XeF ₄

A CONSEQUENCE WORTH QUOTING

Greater **s character** means the hybrid orbital is held closer to the nucleus, so the bond is **shorter** and the atom behaves as more electronegative. This is why C-H bond length falls in the order sp³ (ethane, 109 pm) > sp² (ethene, 107 pm) > sp (ethyne, 106 pm), and why ethyne is acidic enough to react with sodium while ethane is not.

METHOD 2 Finding the hybridisation of any central atom

1. Compute the steric number $H = \frac{1}{2} [V + M - C + A]$, where V = valence electrons of the **central** atom, M = number of **monovalent** atoms attached (H, F, Cl...), C = charge if a cation, A = charge if an anion.
2. Read off: $H = 2 \rightarrow \text{sp}$, $3 \rightarrow \text{sp}^2$, $4 \rightarrow \text{sp}^3$, $5 \rightarrow \text{sp}^3\text{d}$, $6 \rightarrow \text{sp}^3\text{d}^2$.
3. Divalent atoms such as O and S attached to the centre contribute **nothing** to M — they take their electrons back through a double bond.

NH_4^+ : $\frac{1}{2}[5 + 4 - 1] = 4 \rightarrow \text{sp}^3$. SO_4^{2-} : $\frac{1}{2}[6 + 0 + 2] = 4 \rightarrow \text{sp}^3$. ClF_3 : $\frac{1}{2}[7 + 3] = 5 \rightarrow \text{sp}^3\text{d}$. XeF_4 : $\frac{1}{2}[8 + 4] = 6 \rightarrow \text{sp}^3\text{d}^2$. CO_2 : $\frac{1}{2}[4 + 0] = 2 \rightarrow \text{sp}$.

GEOMETRY IS NOT THE SAME AS SHAPE

The steric number gives the **electron-pair geometry**. The **shape** is what remains when the lone pairs are made invisible. H_2O is sp^3 with a *tetrahedral geometry* but a **bent shape**; XeF_4 is sp^3d^2 with an *octahedral geometry* but a **square planar shape**. Answer whichever the question asks for, and name the other to be safe.

6 · Molecular Orbital Theory

Atomic orbitals of the combining atoms merge into **molecular orbitals** that belong to the molecule as a whole. Two atomic orbitals combining by **LCAO** always give two molecular orbitals: one **bonding** (lower energy, additive combination, electron density between the nuclei) and one **antibonding** (higher energy, subtractive, with a node between the nuclei).

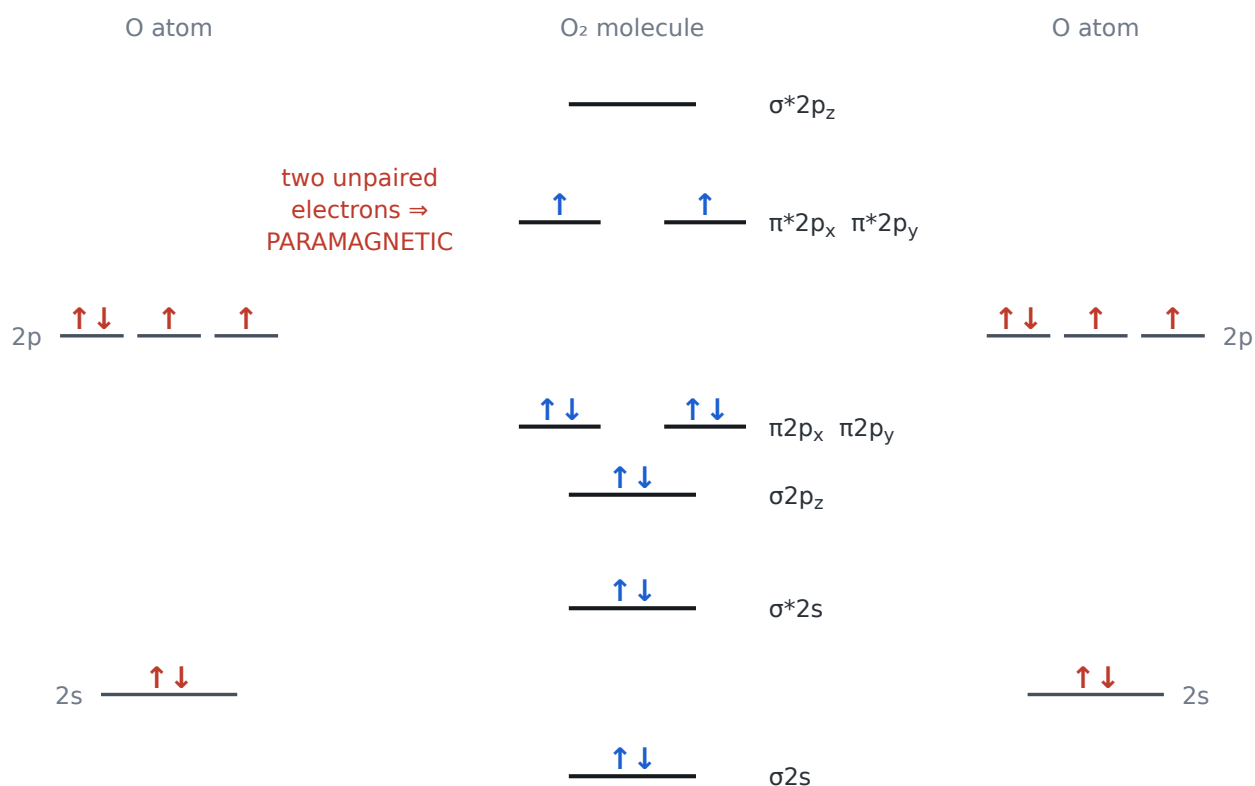
Conditions for combining: the atomic orbitals must have (i) **comparable energies**, (ii) the **same symmetry** about the molecular axis, and (iii) **sufficient overlap**.

Energy order.

For B_2 , C_2 , N_2 (**14 electrons or fewer**): $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

For O_2 , F_2 , Ne_2 (**more than 14**): the two swap — $\sigma 2p_z < \pi 2p_x = \pi 2p_y$.

Reason: in the lighter atoms the 2s and 2p levels are close enough to mix, and s-p mixing pushes $\sigma 2p_z$ above the π orbitals.



$N_b = 8, N_a = 4$ (valence orbitals) $\Rightarrow$ bond order $= (8 - 4)/2 = 2$, a double bond

MO diagram for O₂ (valence orbitals; the 1s electrons are written KK). The two unpaired electrons in the degenerate π* orbitals — placed there by Hund's rule — explain the **paramagnetism of oxygen**, which Lewis and valence bond theories both get wrong. For N₂ the π2p and σ2p levels are the other way round.

Bond order $= \frac{N_b - N_a}{2}$, where N_b and N_a are the numbers of bonding and antibonding electrons. A **positive** bond order means the molecule exists; **zero or negative** means it does not. Bond order $\uparrow \Rightarrow$ bond enthalpy $\uparrow$ and bond length $\downarrow$.

Species	Valence electrons	Bond order	Magnetic behaviour	Comment
H ₂	2	1	Diamagnetic	$\sigma 1s^2$
He ₂	4	0	—	Does not exist
Li ₂	2 (valence)	1	Diamagnetic	Exists in the vapour
Be ₂	4	0	—	Does not exist
B ₂	6	1	Paramagnetic	One electron in each of $\pi 2p_x, \pi 2p_y$
C ₂	8	2	Diamagnetic	Both bonds are π — no σ from 2p
N ₂	10	3	Diamagnetic	Shortest, strongest — 946 kJ mol^{-1}
O ₂	12	2	Paramagnetic	Two unpaired π^* electrons
F ₂	14	1	Diamagnetic	Weak single bond
Ne ₂	16	0	—	Does not exist

WORKED EXAMPLE 2 — THE OXYGEN FAMILY, A FAVOURITE ORDERING QUESTION

Adding electrons to O₂ puts them into **antibonding** π^* orbitals, so each one lowers the bond order:

Species	O ₂ ⁺	O ₂	O ₂ [−]	O ₂ ^{2−}
Valence electrons	11	12	13	14
Bond order	2.5	2	1.5	1
Unpaired electrons	1	2	1	0

Bond length: $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$. **Stability** runs the other way. **O₂^{2−} is the only diamagnetic member.**

7 · Hydrogen Bonding

A **hydrogen bond** is the electrostatic attraction between a hydrogen atom covalently bonded to a highly electronegative, small atom (**F, O or N**) and a lone pair on another such atom. Strength $10\text{--}40 \text{ kJ mol}^{-1}$ — far weaker than a covalent bond (~ 400) but far stronger than ordinary van der Waals forces.

	Intermolecular	Intramolecular
Between	Two different molecules	Two groups within the same molecule
Effect	Associates molecules $\Rightarrow$ raises melting point, boiling point, viscosity and solubility in water	Ties the molecule up internally $\Rightarrow$ lowers boiling point and water solubility
Examples	H ₂ O, HF, NH ₃ , alcohols, carboxylic acid dimers, p -nitrophenol	o -nitrophenol, salicylaldehyde, salicylic acid

THE TWO ANSWERS THIS ALWAYS DEMANDS

Why is ice less dense than water? In ice each oxygen is tetrahedrally hydrogen-bonded to four others, producing an **open cage-like structure** with large empty spaces. On melting, some of these hydrogen bonds break, the cage collapses and the molecules pack closer — so water at 273 K is denser than ice, and ice floats.

Why does o-nitrophenol boil lower than p-nitrophenol? The *ortho* isomer forms an **intramolecular** hydrogen bond (the groups are adjacent), so its molecules are not associated with one another and it is volatile. The *para* isomer can only form **intermolecular** hydrogen bonds, associating the molecules and raising the boiling point — and making it far more soluble in water.

THE SUBTLE ORDERING

Hydrogen bond **strength** follows electronegativity: **HF > H₂O > NH₃**. But boiling point follows **H₂O > HF > NH₃**, because each water molecule has **two** hydrogens to donate *and* **two** lone pairs to accept, so it forms about four hydrogen bonds, whereas HF has only one hydrogen to donate. Total hydrogen bonding beats individual strength.

8 · Where marks are lost

THE TWELVE TRAPS OF CHAPTER 4

1. **Geometry $\neq$ shape.** H_2O : tetrahedral geometry, bent shape. XeF_4 : octahedral geometry, square planar shape.
2. **Lone pairs go equatorial in a trigonal bipyramid** and *trans* in an octahedron. This is why SF_4 is a see-saw and XeF_4 is square planar.
3. **NF_3 has a smaller dipole moment than NH_3** — bond dipoles oppose the lone-pair dipole in NF_3 and reinforce it in NH_3 .
4. **A molecule with polar bonds can be non-polar** if it is symmetrical: CO_2 , BF_3 , CCl_4 .
5. **Resonance forms do not interconvert** and do not exist separately; nuclei stay put, only electrons are redistributed.
6. **O_2 is paramagnetic** and only MOT explains it. Do not answer this with a Lewis structure.
7. **The MO order changes at 14 electrons.** Use π below σ_{2p} for B_2 , C_2 , N_2 ; σ_{2p} below π for O_2 , F_2 .
8. **Adding electrons to O_2 weakens it**, because they enter antibonding orbitals.
9. **Formal charge is bookkeeping**, not a real charge, and the formal charges must sum to the ion's overall charge.
10. **Bond angle order $\text{NH}_3 > \text{NF}_3$ and $\text{NH}_3 > \text{PH}_3$** — electronegativity of the surrounding atom, and size of the central atom, respectively.
11. **Hybridisation does not need half-filled orbitals**, and the number of hybrid orbitals always equals the number of atomic orbitals mixed.
12. **An intramolecular hydrogen bond lowers the boiling point**; only the intermolecular kind raises it.

Chapter 4 of the rationalised NCERT (2023–24 onward). Bond enthalpies in kJ mol^{-1} , bond lengths in pm, dipole moments in debye.